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REVIEW PAPER

A survey on information diffusion and competitive influence maximization in social networks

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Abstract
Social Networks consist of nodes representing users and edges representing interaction among them. They are widely used platforms for information diffusion among various nodes, and influential nodes play a vital role in quickly spreading messages. In viral marketing, information diffusion means to spread information about products, events, and many more to achieve maximum information spread. One of the challenges of information diffusion on social networks is influence maximization, which deals with identifying an optimal number of nodes whose collective influence spread in a social network is maximum. Researchers studied it at an overall network or a community level and as a Competitive Influence Maximization problem where more than one competitor is involved in the information diffusion process. The main highlights of this survey paper are related to various concepts required to be understood in the field of information diffusion and its application in Competitive Influence Maximization. This survey paper focuses on information diffusion and its applications, major factors affecting it, and research challenges in this domain. It also covers the essential concepts of influence maximization and its different types. In addition, variants of competitive influence maximization studied by various researchers in the past two decades are also given. The survey paper also covers various information diffusion models and their extended versions for competitive scenarios, datasets used, and challenges ahead in the Competitive Influence Maximization research domain.

Keywords Social networks · Information diffusion · Influence maximization · Competitive influence maximization · Information diffusion models · Competitive influence maximization algorithms

1 Introduction

Social Networks (SNs) are represented using graphs having nodes and edges where nodes are actors or people and links (edges) represent communication or interaction between nodes, as shown in Fig. 1a and b. Labels on edges indicate a relationship between people (Li et al. 2018a). Social Networks Analysis (SNA) is an interdisciplinary research field primarily rooted in social science, mathematical concepts, graph theory, and computer science. Social Networks come under complex networks; other examples of complex networks are biological networks, transportation networks, and WWW. Table 1 gives details of nodes and edges corresponding to examples of complex networks in different domains. Online Social Networks (OSNs) such as Twitter, Facebook, Instagram (Pham et al. 2019a), and co-author networks are helpful for communication among friends, colleagues, organizations, etc. Users on the network can share an image, video, audio, or text message regarding any event, product, or innovation. OSNs are also used for data collection and analysis for different purposes, such as opinion leader mining, sentiment analysis, community detection, trending topic detection, link prediction and generating recommendations by tracking user behavior. Identifying unusual or fraudulent behavior such as bots, spammers, or fraudulent accounts, is essential for maintaining the integrity of social networks. Applications

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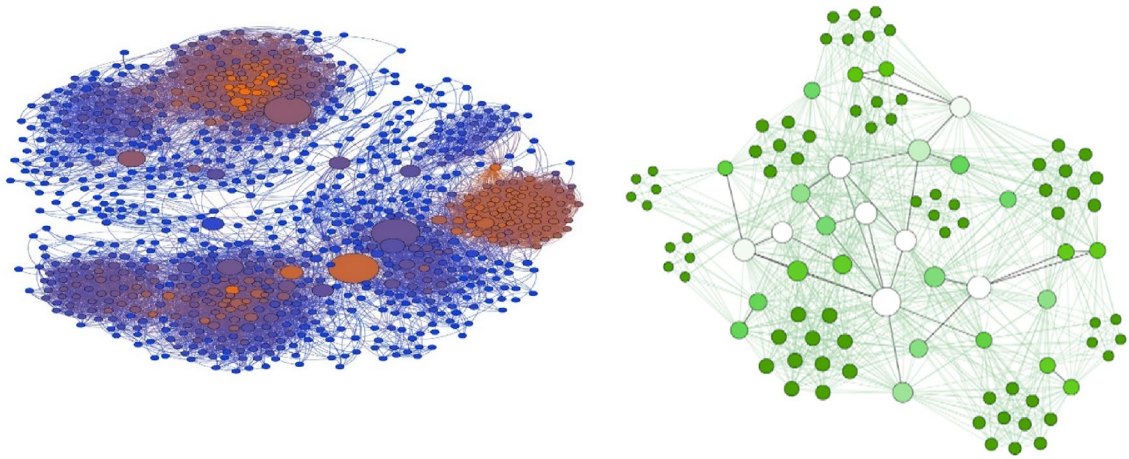


(a) Example of Complex Network1 **(b) Example of Complex Network2**

Fig. 1 **a** Example of complex network1, **b** Example of complex network2

Table 1 Examples of complex networks

Name of complex network	Nodes	Edges	Type of network
World Wide Web	WWW documents	URL links	Scale-free networks
Scientific collaborations	Authors / Scientists	Writing research papers jointly	Scale-free networks
Metabolic networks	Chemicals (Substrates)	Bio-chemical reactions	Small world and Scale-free networks
Yeast protein interactions	Proteins	Physical interactions	Scale-free networks
Electrical power grid	Generators/ Transformers	Transmission lines	Small world networks
Neural networks	Neurons (Cells)	Synapses (Nerves)	Small world networks

of OSNs extensively use machine learning algorithms and deep learning approaches like graph neural networks. Social networking sites perform various functions like instant messaging, media sharing, microblogging, social networking, and many more. Social networks are extensively used for information dissemination (diffusion) to many people, using different propagation models and approaches in less time. In addition to online marketing (viral marketing) related to various goods and services, information diffusion (ID) can also be used for sharing an opinion or feedback on different matters of public interest, running awareness campaigns related to social issues, identifying breaking news, holding online election campaigns, blocking rumors, and handling emergencies due to various disasters.

In the past few years, the use of OSNs like Facebook, Flickr, Twitter, etc., has increased. Researchers conducted research in areas like ID to know the source or initiators of the diffusion process in the network, and nodes played a crucial role in information spread along with their influence spread computation. The problem of Influence Maximization (IM) refers to selecting a set of seed nodes so that the total influence spread can be maximized (Li et al. 2018a) in non-competitive and competitive scenarios. In recent years, influence maximization has been researched in competitive scenarios where more than one competitor is trying to diffuse information about their product to influence the maximum number of people on the network compared to their competitors (Pham et al. 2019a).

CIM was first addressed by Bharathi et al. using a $(1 - 1/e)$ Approximation algorithm (Bharathi et al. 2007). Bozorgi et al. studied the CIM problem using the "followers perspective," and they tried to achieve maximum influence spread by choosing a minimum number of seed nodes for competitor 2 with pre-knowledge of seed nodes assigned to competitor 1 (Bozorgi et al. 2017). W. Lu et al. solved CIM from the "host perspective", that is, the owner of the OSN allocates seed nodes to all competitors fairly as per their budget to maximize collective spread and maximum benefit to each competitor that defines value for their money (Lu and Bonchi 2013). A. Masucci et al. worked on the strategic resource allocation of competing marketing campaigns such that it can be decided on how many resources are to be allocated to potential customers who can advertise their products using game theory and voter model (Masucci and Silva

2014). A. Ansari et al. also worked on a similar problem in competitive influence propagation where the main focus was to allocate resources to influential nodes to advertise their products with a difference that their action space was discrete rather than continuous, as in earlier studies related to budget allocation (Ansari et al. 2019). H. Li et al. solved the CIM problem as a "Dominated Competitive Influence Maximization" (DCIM), where the focus was to maximize the difference between the influence spread of the dominating competitor over other competitors using the CIC-M model (Li et al. 2018b). Y. Dai has solved CIM as a GCIM problem for group-based influence maximization using the CG-LT model (Dai 2018).

C. V. Pham et al. studied the CIM as a "Budgeted Competitive Influence Maximization (BCIM) problem" under budget and time constraints. They also assumed fair competition among competitors using an approximation algorithm based on the "Sandwich framework and polling-based method" (Pham et al. 2019b). Another work by C. Pham et al. studied the CIM problem as a deterministic information propagation using the DCLT model, in which two separate thresholds are assigned to each node for two different competitors to achieve diffusion. S. Gao et al. addressed CIM as a novel "Fair-aware Competitive Event Influence Maximization (FCE-IM)" problem using the E-LT model for event-based social networks (Gao et al. 2020). Z. Liang studied CIM as a Targeted Influence Maximization (TIM) problem using a "Reverse Reachable set-based Greedy (RRG) algorithm" such that a particular group of users can be recommended with competitive products of various competitors, and a pruning strategy was designed to enhance the performance of the algorithm proposed by the authors (Liang et al. 2023).

The various variants of CIM have been studied to cover different aspects of competitive viral marketing. This field still has the scope to improve the competitive diffusion modeling to get maximum spread. The large-scale and continuously evolving online social networks put before researchers the urgent need to handle the diffusion process efficiently under time and budget constraints for multiple competitors. Table 2 includes abbreviations and their detail used in this survey paper.

This survey paper consists of sections, as shown in Fig. 2: Sect. 2, which focuses on related surveys. Section 3 highlights the motivation behind this survey paper. Section 4 provides the details of sources of information. Section 5 highlights the Introduction to ID and its application areas. Section 6 highlights the various factors affecting ID in SN. Section 7 explains the Research Challenges of ID. Section 8 gives basic terminology for IM. Section 9 explains the types of IM. Section 10 gives a brief introduction to CIM and its applications. Section 11 introduces ID models for competitive and non-competitive IM. Section 12 covers different variants of CIM studied by various researchers. Section 13 focuses on related work on CIM. Section 14 gives details of datasets that are used in CIM-related research works. Section 15 concludes the existing survey and highlights the challenges ahead in CIM.

The major contributions of this survey paper are:

- It presents a detailed overview of different application areas of ID.
- It covers the various factors that affect ID.
- It highlights the existing challenges of ID, including IM, which is further surveyed in detail.
- It covers a detailed overview of different algorithms, models, and datasets used for CIM problem solutions.
- The conclusion is drawn around the existing challenges of CIM at the network and community level.

2 Related surveys

The surveys by various researchers in ID and IM covered ID models and algorithms, influence dynamics, and challenges in the field of IM. Some of the related surveys done in this area are:

The survey in Sun and Tang (2011) focused on computational aspects of the research on social influence analysis, including metrics for measuring social influence, a literature survey on social similarity, and influences. It also covers research on social influence maximization, including techniques and real-time applications, namely online advertisement through viral marketing and prediction of customer behavior. The survey aimed at measuring the influence levels, qualitatively or quantitatively, for the given nodes and edges in the SN.

The researchers in Jayamangala and Sheshasaayee (2015) explored approximation algorithms and diffusion models to achieve IM, and the existing work considered mainly the degree and structure of the network for finding influential nodes in an SN, the number of nearest neighbors, and the connectivity of users based on ratings given by users. They discussed the most popularly used influence diffusion models like the "Independent Cascade Model", "Linear Threshold Model", "Heat Diffusion Model", and an extension of the IC model called the "LAIC (Latency Aware Independent Cascade) Model", which handles influence propagation delay. In addition to models, approximation algorithms like greedy algorithm, greedy hill-climbing algorithm, degree discount heuristic approach, LDAG algorithm for influence maximization, k-step greedy, top-k, and improved k-step greedy algorithms based upon above models of information diffusion are also discussed regarding their advantages and disadvantages.

Table 2 List of abbreviations and their details

Sr. No	Abbreviation	Detail of abbreviation
1	BCIM	Budgeted Competitive Influence Maximization
2	CAIM	Context-aware Influence Maximization
3	CCIC	Competitive and Complementary Independent Cascade
4	CELF	Cost-effective Lazy Forward
5	CG-LT	Competitive Group Linear Threshold Model
6	CI2	Competitive Influence Improvement
7	CIC-M	Competitive Independent Cascade Model
8	CIM	Competitive Influence Maximization
9	CLT	Competitive Linear Threshold
10	CPro	Competitive Profit Maximization for the host
11	CT	Continuous-time
12	DCIM	Dominated Competitive Influence Maximization
13	DCM	Decidable Competitive Model
14	DIC	Dynamic Independent Cascade
15	DT	Discrete-time
16	IC	Independent Cascade
17	ID	Information Diffusion
18	IM	Influence Maximization
19	IS	Influence Spread
20	K-LT	Extended Linear Threshold
21	LT	Linear Threshold
22	SN	Social Networks
23	STM	Separated-Threshold Model
24	TCLT	Time constraint Competitive Linear Threshold
25	VM	Viral Marketing
26	WC	Weighted Cascade
27	WPCLT	Weight-proportional Competitive Linear Threshold Model

In the research paper (Arrami et al. 2017), the authors have given the definition and role of opinion leaders and how centrality and influence maximization techniques can be utilized to identify influential nodes. They also conducted a comparative study between different centrality techniques and provided the limitations of each. It was observed that except for "Degree Centrality," the rest of the centrality techniques require knowledge of the complete network structure.

In (Banerjee et al. 2020), the authors studied the problem of Target Set Selection (TSS), also known as Social Influence Maximization (SIM) in SN. The primary focus of the survey paper is on providing details of different variants of SIM problems, such as finding the Top-k Node Problem, Lambda-Coverage Problem, Budget Influence Maximization Problem, Weighted TSS Problem, and Influence Spectrum Problem. The authors have also developed a taxonomy for various solution methodologies for solving SIM problems under Approximation Algorithms, Heuristic Solutions, Community-based Solution Frameworks, Meta-heuristic Approaches, and other miscellaneous approaches like data-based, hop-based, and query-based. Further, the authors gave future directions like expanding the solution to large-scale networks i.e. scalability issue using parallel and distributed algorithms, computation of diffusion probability on edges, solution of TSS problem under cost, benefit and time constraints, SIM problem solution for time-varying social networks and using models different from basic Linear Threshold and Independent Cascade. The authors have also suggested that the SIM problem can also be worked out under different variants, such as topic-aware influence maximization, and other meta-heuristic approaches can also be tried to find a solution. It has also been observed that the diffusion process needs to be studied under multiple phases rather than a single phase to increase the influence spread.

In (Han and Li 2018), the authors discussed the Social Influence (SI) Analysis covering the development process of influence analysis, description of the classical models for simulation of the influence spreading progress, influence diffusion models, and approaches for IM. In addition, the applications based on influence analysis are discussed, such as marketing, advertisement, and web services. Ultimately,

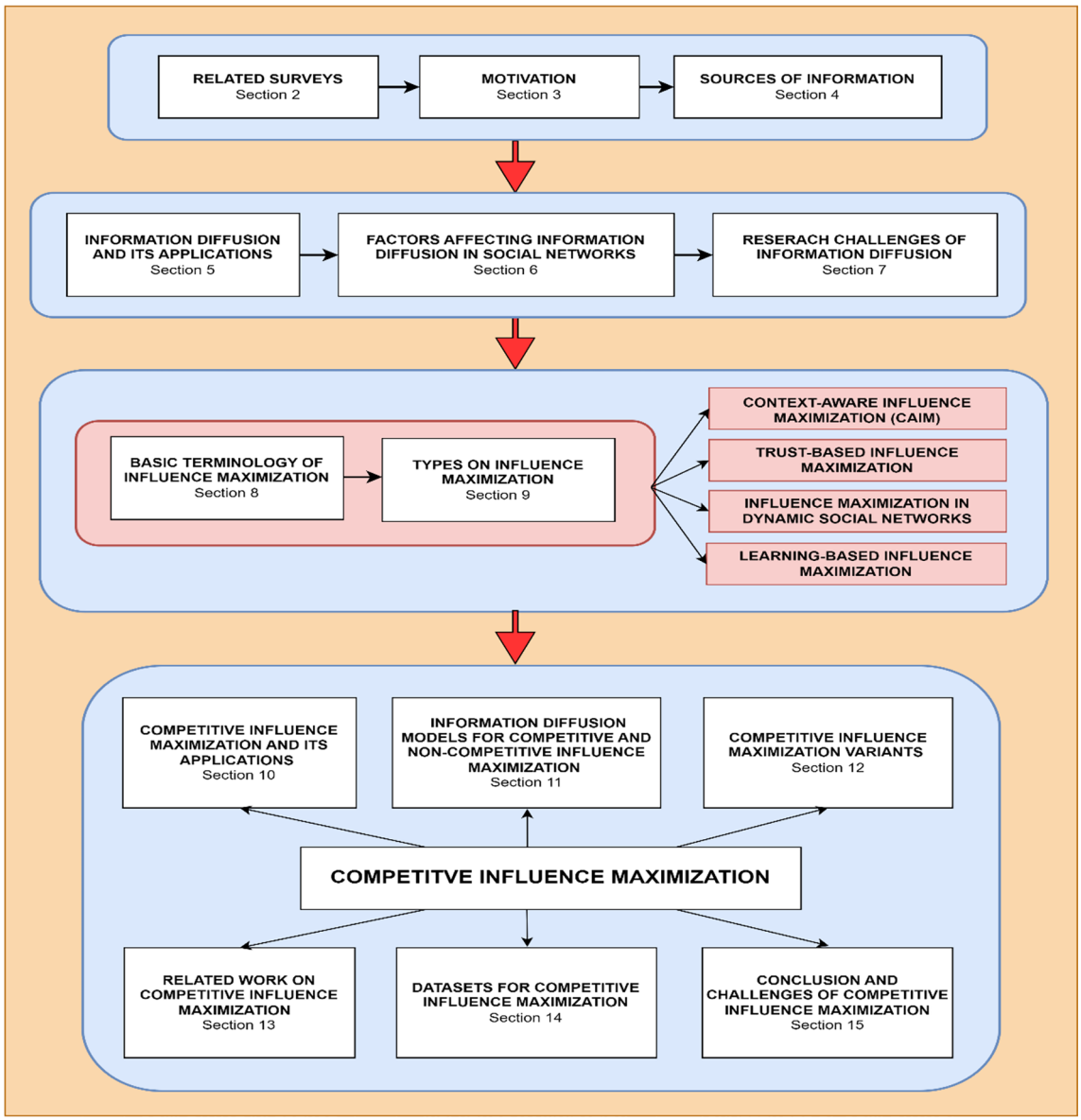


Fig. 2 Flow of survey paper on ID and CIM in social networks

the authors have given research challenges and future directions in this area relevant to academia and industry.

In (Peng et al. 2018), the authors provided a social network overview, including definitions and types of SN, such as topology-based and function-based. In addition, the survey is also focused on covering the understanding of SI analysis concerning its definition, properties, application areas, architecture, various diffusion models, and the various evaluation metrics. The authors have also summarized the existing SI evaluation models, provided an overview of the existing IM methods, discussed the challenges of current algorithms, and discussed future possibilities for research in IM.

In the survey paper (Li et al. 2018a), the authors have covered IM from the point of view of algorithms and highlighted mainly popular ID models used for the IM problem, comparison based on theoretical explanation of algorithms, and taxonomy for the classification of IM algorithms according to their design objectives. In addition, the applications of IM techniques in SN in the context of topic, location, and time are also covered, namely topic-aware IM and location-aware IM. The authors have also discussed the major challenges and future research possibilities.

In the survey (Bamakan et al. 2019), the authors have classified techniques for opinion leader identification(OLI) into descriptive, diffusion process, and hybrid content mining-based approaches, "statistical and stochastic methods", "topology-based methods", data mining, and

learning methods. The pros and cons of each method for OLI are analyzed and compared to provide details about existing research challenges in this field and future trends.

In (Bian et al. 2019), the authors have mainly discussed the two categories of influential node identification, namely "top-k influential" and "top-k significant nodes" in an SN.

In the survey (Singh et al. 2022), the researchers compared the approaches and algorithmic frameworks for IM algorithms. The authors have also discussed and compared the characteristics of various influence diffusion models like Threshold Models, Cascade Models, Triggering Models, Epidemic Models, and Time-aware Models. The authors have also elaborated on terminology related to influence maximization and various challenges in this research field, like designing efficient algorithms for maximizing influence on large-scale social networks. The survey also includes various performance analysis metrics for different influence maximization algorithms, like the efficiency of algorithms in terms of quality of seed set selection, time consumption, scalability, and study of robustness (no change in optimal solution) of IM algorithms with change in influence propagation process.

In the survey (Li et al. 2023), the researcher studied the ML-based solution for IM and its variants in the past few years. ML-based solution combinatorial optimization problem (IM) has resulted in faster solutions and better generalization of unknown graphs. The Deep Enforcement Learning (DRL) framework achieves better learning from graphs and decision-making. Pointer Networks and Graph Neural Networks use DRL to solve IM, and Graph Embeddings are also used. ML-based methods can be categorized into "supervised," "unsupervised or self-supervised," "semi-supervised," and "Reinforcement Learning (RL)," and out of all these approaches, the RL approach for IM is preferable. Due to the characteristics of IM, such as multi-constraints, multi-objective, and non-static, the traditional methods for solving IM problems are not so successful, and hence ML-based approaches are used for solving IM problem. The various challenges of solving IM problems using ML-based methods include scalability, generalizability, training efficiency, diverse diffusion models, etc. Hence, these challenges demand new creative ML-based approaches.

In the research study (Ye et al. 2022), traditional diffusion models are first reviewed, and a comparison of their features was done to provide basic knowledge of the diffusion process. Further, extensions of basic diffusion models regarding context were discussed, and their characteristics were compared. In addition, "deep-learning-based" simulations of diffusion processes have also been summarized. Classic IM algorithms and CAIM algorithms were analyzed and compared, key challenges and future possibilities for working in this domain were given for the benefit of researchers. The research paper also focused on comparing IM algorithms under four different approaches: "simulation-based," "heuristics-based," "sampling-based," and "meta-heuristic-based" under classic diffusion models. It also compared the CAIM algorithms under "context-aware" diffusion models.

3 Motivation

There are very few surveys done that are based explicitly on Competitive Influence Maximization (CIM) and various aspects related to it, such as models that can be used for diffusion of information in a competitive scenario, which datasets are suitable for CIM work, approaches for solving competitive influence maximization, and challenges ahead. The domain of CIM got the attention of researchers due to its applicability in viral marketing. The survey provides all the relevant information to the researchers required to work in this domain. In addition, community-based CIM is a step towards reducing the time needed to diffuse the information in the network, so existing research on community-based CIM is also discussed in detail.

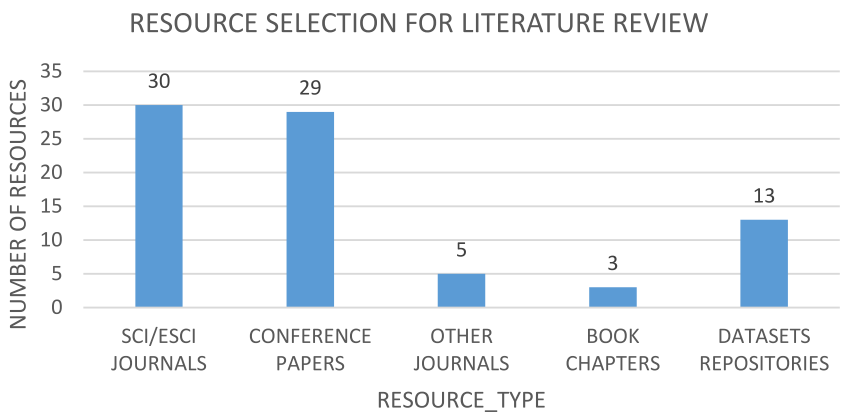
4 Sources of information

In this survey paper, various resources were referred to, including research papers from reputed SCI Journals, International Conference Proceedings, Book Chapters, and Web Resources for exploring Data Repositories for Online Social Networks, as shown in Fig. 3a.

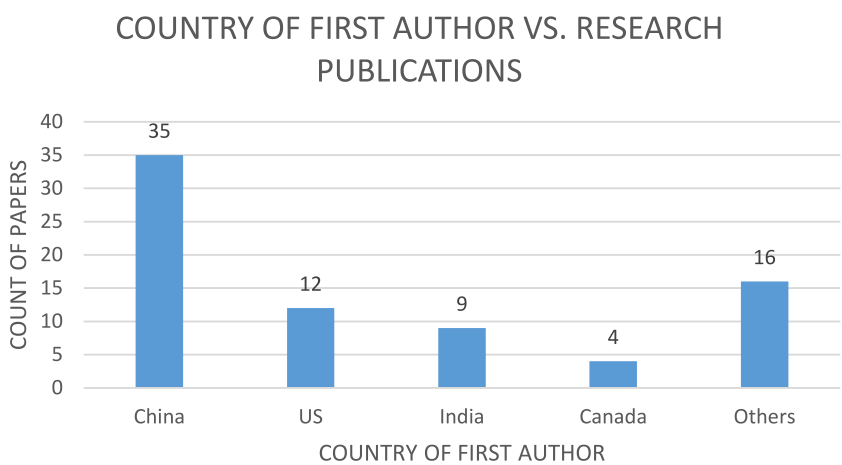
The publications referred to in this study are mostly authored by researchers from countries like China, the United States, India, and Canada. Still, a few CIM papers are available from countries such as Africa, Australia, France, Greece, New Zealand, Poland, Romania, Singapore, the UK, and Vietnam, as shown in Fig. 3b. So, a survey paper has been written, considering the diversity in work done by various researchers on CIM.

5 Information diffusion and its application areas

Information diffusion refers to the adoption of some information received from its contacts (neighboring nodes) on a social network by getting influenced by them. Social network users diffuse information in their contacts by performing user actions like retweets, comments, and mentions on social networks like Twitter (Bharathi et al. 2007). Information diffusion also occurs in user actions, such as liking a post, forwarding the information, and marking something favorite. These user actions help identify user interests/preferences



(a): Resource selection for literature review based on type of publication



(b): Number of research publications from different countries based on country of first author

Fig. 3 **a** Resource selection for literature review based on type of publication, **b** Number of research publications from different countries based on country of first author

and contribute to finding an individual's probability of propagating information in an online social network.

Various application areas of ID include fields like viral marketing, breaking news detection, election campaigns, rumor blocking (Bozorgi et al. 2017), disaster management in emergencies (Lu and Bonchi 2013), etc. This section explains ID applications, with special emphasis on their applicability in CIM scenarios. The application areas of ID are shown in Fig. 4.

5.1 Viral marketing

Viral Marketing is all about using internet-based means like e-mail communication, forums, blogs, and social networking websites, etc., to advertise content, promote awareness about their brand, and spread information about some new product or innovation (Masucci and Silva 2014) across different countries or within targeted customers or online users.

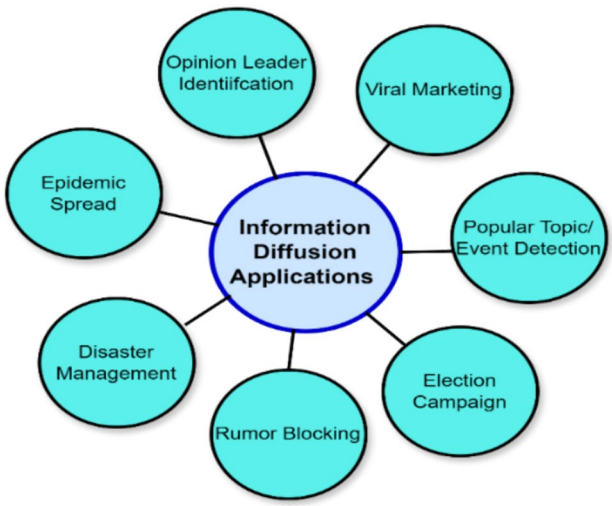


Fig. 4 Applications of information diffusion

Traditional advertising approaches, like radio, TV, printed material, etc., are pretty costly, and their reach is limited, and there is a time constraint also.

Content in the form of images, videos, memes, and articles goes viral on the internet when the matter interests the user, is entertaining, thought-provoking, valuable, contains some shocking message, or has urgent information to share. Social network viral marketing has the added advantage of advertising to target users or people at a large scale within less time, using less cost. Moreover, the content is available for access and sharing among people in your contact list anytime and anywhere using various platforms like desktop, laptop, and mobile-based internet access. Viral marketing-based research makes use of influence models that satisfy properties like submodularity (mandatory) and incrementality property (not required) (Ansari et al. 2019). Viral marketing campaigns are either user-generated or, with influencers' help, the content is spread using planned strategies. Zoom's Virtual Background Challenge is one of the most successful viral marketing campaigns. During the COVID-19 pandemic, the demand for work from home and the use of virtual meetings increased to a great extent. Due to this prevailing trend, Zoom initiated the Virtual Background Challenge to encourage users to share creative and fun virtual backgrounds during Zoom meetings. Zoom used hashtags and promoted its campaign on social media to attract new customers, interact with its current users, and reach more people. Another viral marketing campaign was run by Spotify. In 2019, Spotify entered the Indian Market with a major Diwali campaign, "Spotify Lights Up Diwali: Celebrating Festivals with Music," starring Bollywood Actors Ishaan Khattar and Anil Kapoor on YouTube advertisement. This advertisement was viewed by 50 million people, showcasing Spotify's enormous song catalog containing more than 50 million tracks.

Viral marketing campaigns based on planned strategies and appropriate social media platforms lead to success and provide broader coverage of their product to online users.

5.2 Popular topic/Event detection

On various social media and content-sharing websites, identifying popular or trending topics is a beneficial application area in ID (Panagiotou et al. 2016). A topic or event is supposed to be in trend if many users concurrently access it over a short period and not before or after that. Trending topics boost rapid ID across social networks as they reflect the number of users interested in them. Event detection has many challenges due to the vast amount of data continuously streamed at high velocity on different social media platforms. Moreover, text mining is a less effective approach for event detection due to short and incomplete messages posted by users (Alkhamees and Fasli 2016). Real-time event detection also becomes tough when events are sudden and unplanned, and the accuracy of event detection highly depends upon a high streaming level. The timeliness of posts also matters when it comes to detecting disaster-related events like earthquakes, wars, etc. There is also a relation between the occurrence of bursty events and the evolution of networks in terms of more nodes connecting the network (Yang and Wu 2022). The frequency of communication is also very high compared to regular events.

5.3 Election campaign

Social networking sites (SNS) provide scope for information dissemination to online users on a wide scale. This understanding made it easier for the various political parties to promote their election campaigns using SNS like X (formerly known as Twitter), Facebook, Instagram, Telegram, and WhatsApp. These have been the most popular social networking platforms among internet users for over a decade, and nowadays, these are extensively used in building opinions about various parties based on messages shared and liked by multiple users (Ansari et al. 2020). These are major sources of big data and are used further for sentiment analysis or opinion poll mining to decide upon various election strategies to be used in the future by different political parties. Moreover, public opinion mining in election campaigns differs from traditional survey data, giving better opportunities to non-active individuals in traditional surveys (Ni et al. 2020). It also takes care of influence due to the opinions of others in the network, and it helps predict information diffusion related to different political parties. The accuracy of prediction may also vary depending on the various methods used for opinion mining. In India, political parties like BJP and Congress also used social media platforms to spread their party's agenda in the 2024 Lok Sabha elections and convince voters to vote for them using creative memes, animated video films, articles, etc.

5.4 Rumor blocking

Online social networks provide quick access to a vast amount of information, like large-scale news data, and help spread information faster, in many cases even quicker than mass media. In addition to this benefit, sometimes they have a devastating effect due to the spread of wrong information such as rumors of earthquakes may create panic among people to rush here and there to protect their lives, spread of misinformation on Twitter like death news of some celebrity, quick hike in the oil prices and hence they are not reliable source of information propagation (Li et al. 2018a). The research on rumor blocking was also done concerning disjoint communities based on identifying nodes, which can

protect or block rumor spread by spreading relevant and correct information in the network (Ni et al. 2020). Efforts have been made to limit the spread of misinformation by parallel propagation of appropriate information in a timely and cost-effective manner (Zhang et al. 2015).

5.5 Disaster management

Social media is considered one of the fastest, most feasible, and most cost-effective sources of communication in times of disaster management and extreme events to reach more people regarding reliable and updated information. Because of this, many government and private agencies use this on a large scale for message propagation. Social media portals play a significant role in spreading current event-related information to individuals, groups, or communities. In this manner, by creating awareness, many lives can be saved, facilitating decision-making to handle alarming situations. Twitter, Facebook, and mobile social networks like WhatsApp are a few of many organizations' most highly used social media sites (Pham et al. 2019a). These platforms are equally important for issuing disaster-related information and effectively handling post-disaster rescue operations (Dong et al. 2018).

5.6 Epidemic spread

Epidemic spread has been studied in correlation with information spread about the disease in complex networks by many researchers by using a two-layered approach (Zhan et al. 2018) (Guo et al. 2021). The studies identified a strong relation between information spreading among influential nodes and its influence on patterns of disease spread. Moreover, this helps prevent disease from spreading in complex networks. On the other hand, the impact of excessive ID related to diseases like coronavirus has increased fear among individuals to a great extent. Further, it leads to a shortage of limited resources due to their hoarding (Wang et al. 2021). This kind of panic situation also resulted in disease spread as more people started rushing to markets to collect essential resources for consumption during the lockdown. In addition, the Government has taken social media-led initiatives using celebrities (influencers) to motivate people nationwide to go for COVID-19 vaccination to control the spread.

5.7 Opinion leader identification

It is a challenging application area of ID or broadly in social network analysis. Influencer or extraordinary individual identification is quite popular for advertising a new product for running awareness and election campaigns. Opinion Leader identification can be classified into different categories (Bamakan et al. 2019): (i) Influence scope classifies into local and global, (ii) Knowledge domain classifies into Monomorphic (having expertise in one domain) and Polymorphic (having expertise in multiple domains), (iii) Behavior classifies into positive and destructive (iv) Time classifies into "short-term" and "long-term." The diffusion-based approach plays a vital role in deciding which node can diffuse to other nodes better in social networks. There are many models to diffuse information, like the LT Model effectively, the IC Model, the "Weight-Proportional Competitive Linear Threshold Model," the separated-threshold Model" for competing technologies, the Decidable Competitive Model propagation, etc. In addition, there are several approaches or methodologies for opinion leader detection, such as descriptive approaches, approaches based on diffusion process, statistical and stochastic methods, topological-based methods, and hybrid content mining-based approaches (Parau et al. 2017). Opinion leader identification is of utmost importance in CIM, where efficient influential nodes or opinion leaders can help achieve company goals within less time by adopting the idea at the initial stage and then spreading it in their network.

6 Factors affecting information diffusion in social networks

Researchers have identified several factors that affect the ID mainly based on network structure, such as tie strength, network density, centrality measures, and homophily in SN or communities (Yang et al. 2013). Further, many factors revolve around users representing nodes in SN, such as opinion leaders (influential nodes) identification, user activation thresholds, user interest/preferences for specific topics or domains (Pham et al. 2019a), user emotion/sentiment analysis, number of user friends, etc. Identifying such factors has become essential for maximizing ID and has made influence analysis a vital research area to bring innovation and efficiency to ID.

The Influence can have characteristics like dynamism, i.e., it may increase or decrease with time, and influence propagates over the SN. Further, influence is transitive and asymmetric, and the acceptability of influence is subjective (Bozorgi et al. 2017). Some of the significant factors which affect ID are:

6.1 Network structure

It varies in terms of nodes and edges count, node degree, and interaction change over time in terms of strength of ties and conversation dynamics. Further, network topology and initial values for network parameters also help control the diffusion intensity and rate (Kumar and Sinha 2021).

Individual nodes representing influential nodes act as a source of motivation for others in the overall diffusion process. Based on network structure, opinion leaders were identified in a social network by computing centrality measures. Researchers have also worked on hybrid centrality measures to analyze the significance of nodes (Ahmad et al. 2020), such as the dynamic weighted sum method based on community structure to identify influential nodes.

Network dynamics refer to the evolving structure and relationships within a network, such as adding or removing nodes (representing individuals or entities) and edges (representing relationships). These changes alter the network's topology in terms of network density, clustering coefficient, and influence pathways, affecting both the diffusion process and the strategies for influence maximization. Here's a breakdown of the key impacts of network dynamics:

- **Influence Spread Variability and sustainability** Dynamic networks change influence paths, affecting reach and effectiveness. In such networks, adaptive influence strategies are required to maintain spread over shifting connections. As influence decay is a risk in dynamic networks, periodic reseeding or repositioning of the influential nodes helps sustain spread. Predictive and Probabilistic Models anticipate future network states, helping identify resilient influence pathways and seed nodes.
- **Temporal Dynamics in Influence Activation** Influence is time-sensitive in dynamic networks, with activation depending on evolving connections. Temporal centrality can help target influence during critical connectivity periods. Moreover, a node's influence in dynamic networks fluctuates over time, necessitating frequent recalculations and predictive approaches to capture future influential nodes.
- **Overlapping Community Influence** Network evolution creates fluid community overlaps, requiring influential nodes to be placed near bridges for cross-community influence.
- **Adaptive Influence Algorithms** Algorithms must adapt to network changes, using reinforcement learning to adjust influence sources as the network evolves continuously. Real-time adaptive algorithms are essential for immediate influence in trending events, where network conditions shift quickly.

6.2 Tie strength and node position

Tie strength means which nodes are connected and how strongly they are connected in an SN. In a Weighted SN, tie strength can be considered as the frequency of communication or strength of influence represented by the influence probability on edges of the social network. Ties can generally be classified as strong and weak ties (G. Mark S. 1973). In SN, strength is not only in strong ties among nodes but also in weak ties. Weak ties are important, especially when establishing collaborations among social networks representing companies or organizations. The node position in an SN plays a significant role in diffusing information, and sometimes, it's a node that connects two different components of a network or two entirely different SNs (Musiał and Juszczyszyn 2009). Border nodes in an SN can also be promising for ID if connected to nodes in other social networks.

6.3 User characteristics and homophily in social networks

Users on social networks can also have attributes assigned to them. User attributes can be expressed in terms of demographic features (age, location, interests, or preferences in a particular context), and they help in finding similarity among adjacent nodes in the SN. Similar characteristics of users in Social Networks positively affect ID as a node will prefer to diffuse information to another node identical to it in interest or preferences (Pham et al. 2019a). The user influence on similar nodes is assumed to be more. Further attributes related to users or information are relevant predictors of diffusion efficiency along with the topology and dynamics of SN.

Homophily in social networks is closely associated with identifying nodes with similar interests or characteristics, which becomes an added advantage for further diffusing information in the network (Salem et al. 2011) (Yang et al. 2013). The social networks exhibiting this property are Facebook and Collaboration Networks, representing facts like friendship, co-authors or co-workers, etc. It is also one of the popular approaches for community detection for quickly diffusing information to a group of nodes in an SN. Further, the network density lies between 0 and 1, and the network with high density leads to quick ID on social networks.

6.4 Centrality measures and opinion leader identification

The measures computed based on the position of the nodes in the SN, called centrality measures like closeness and degree centrality, were also used by many researchers to predict information diffusion through the identification of influential nodes (Musiał et al. 2009). Centrality measures represent a person/actor's importance on the SN and how much they can contribute to ID. Each measure is used in different situations and has diverse applications and computational complexity.

Identification of opinion leader(s) can play a significant role in ID because they have more potential for influencing

others, and influenced individuals may further propagate the information in their influence circle on the social network. Centrality measures such as degree centrality and clustering coefficient can be used to mine opinion leaders from online SN (Srinivas and Velusamy 2015). The influence of an individual may also vary over time, and it directly impacts opinion leader detection in online SN (Arrami et al. 2017) (Bamakan et al. 2019) (Parau et al. 2017). Further, it will affect the information diffusion cascade probability.

6.5 Influence diffusion probabilities

In ID models like the LT model, every node in the SN has a threshold value between [0,1], and each edge in the SN has an influence probability, which helps in deciding whether a particular activated node can influence neighboring nodes. The influence probabilities can be assigned equally on all edges, randomly assigned, or calculated using different approaches like the Bayesian networks-based approach (Varshney et al. 2017). Influence probabilities among users change over time and are associated with the strength of ties in OSN. The topology of the SN is insufficient to compute the information diffusion probability, so many researchers have tried homophily-driven diffusion. In such cases, diffusion is highly affected by the similarity of interests between users of OSN (Lagnier et al. 2013). Similar user profiles indicate common preferences or interests, and it is a highly promising situation for ID or, in the case of product adoption, the probability of accepting or buying a product increases.

6.6 User emotion analysis/Sentiment analysis

In social networking sites like Twitter and other micro-blogging sites, sentiment analysis of different actors (nodes) on the network can be done to analyze the probability of information diffusion. Further opinion or sentiment evolves in SN, which may affect the influence of a node. Further opinion evolution leads to faster ID if diffusion is intervened by one-sided opinion (Zhu et al. 2018). Earlier researchers have focused on ID and opinion evolution as two processes that are independent in nature rather than studying the interaction between them. Due to this, the researchers tried to find a correlation between ID and opinion evolution on SNs and quantified sentiment's effect on ID (Ferrara and Yang 2015). Negative messages spread faster, but it has been found that the reach of positive messages in the network is comparatively greater, and it further signifies the intention of online users to share and propagate positive messages to more and more persons for a longer duration.

6.7 Social influence and personal preference

Social influence can be measured using various performance metrics like centrality measures and frequency of interaction between users (Bozorgi et al. 2017). This process generally involves two entities, namely, influencer and influencee, and the probability with which the influencer can influence the other person in the network is uncertain. This uncertainty is expressed from strong to weak by assigning a numerical value representing the probability of influence on the edges between different users in the network. Social influence is an asymmetric relation.

6.8 User login frequency

User login frequency signifies the speed of information diffusion in SN. Passive users do not contribute much towards ID as their login frequency is low, which may delay the time to propagate the information further to the connected users in their SN (Li et al. 2017). So, to speed up ID, the login frequency of users can also be considered a criterion for selecting nodes to diffuse information further. Information diffusion history tells about who diffuses information to whom and at what time.

6.9 User influenceability and activation threshold

The vulnerability of a node can be expressed as the probability of getting influenced by other nodes, and it helps predict whether an individual will propagate information further or not. The probability of getting influenced may vary due to factors like activation threshold, personal preferences or opinions about different influencers, etc. If the total influence of incoming edges at a specific node exceeds its threshold limit, the node will be activated; otherwise, it will not. Further, influence exhibits certain properties like subjective, transitive, dynamic, propagative, and measurable (Peng et al. 2018).

6.10 Incentive scheme

Some incentive schemes can also be used for individuals, who agree to propagate information to increase diffusion probability. This kind of work requires payoff calculation and is supported by game theory as well for deciding to diffuse information further.

The above factors play a vital role in the prediction of ID, and influence analysis is one of the recent research areas in ID.

7 Research challenges of information diffusion

Information diffusion in SN has been widely studied by researchers in the past, and the various challenges highlighted by them are given below:

7.1 Modeling information diffusion over dynamic social networks

Information propagation is associated with the activation sequence of nodes and the time at which information is diffused. Usually, the models used for capturing and identifying mechanisms for ID make use of two types of models, namely Explanatory Models (ExM) and Predictive Models (PrM) (Bharathi et al. 2007). The ExM are used for understanding how information propagated in OSN or which path consisted of an activation sequence of nodes was taken to diffuse information. The PrM emphasizes how the information will be propagated based on known diffusion history. PrM are categorized as graph-based and non-graph-based approaches. LT and IC Models fall under the graph-based predictive models category. Both the models make use of static graph (directed) structure.

7.2 Prediction of temporal dynamics of information diffusion

Information diffusion prediction in OSN is correlated with predicting user behavior and influence analysis. It is highly dependent on network dynamics and time-dependent, that is, temporal. It varies depending upon factors, directly or indirectly, such as opinion evolution, network structure, individual influence and influence probability, personal preferences, and trust propagation, which is also related to information diffusion.

7.3 Node label prediction and node classification problem

Node labels represent demographic information (like gender, age, occupation, location) and interests/hobbies (movie watching, music, book reading, etc.). These node labels can have one value or multiple values varying from one person (individual) to another and can be static or vary with time. Node labels can also have weighted values to represent some scenarios, like the number of times a video or photograph is tagged. It is pretty challenging to maintain the updated node label information. Node labels are generally assigned by some expert, and in an SN, labels of only a few nodes are often available. So, it is one of the research challenges in SN to predict the label of a node, which is further used to solve a node classification problem. Labels of nodes can also be predicted by using information from neighboring node labels. Local or iterative classifiers and Random walk-based methods can be used for node label identification.

7.4 Influence maximization problem

Research on IM aims to find a subset of nodes called seed sets or initial candidate nodes for spreading information to a maximum number of users within budget constraints on the social network. So, influence maximization depends on how effectively influential nodes can be identified for diffusing information on SN, and further, it also involves in-depth user behavior and influence analysis.

7.5 Network community evolution analysis

Information diffusion highly depends upon how the social network is structured, and it also helps identify a strongly connected group of nodes called communities. Community structures help spread information within that group of nodes compared to nodes outside the community. Hence, this field has high potential in terms of diffusing information quickly to nodes within the community.

7.6 Ethical considerations of information diffusion

In social media (digital) marketing, the primary issue is maintaining transparency in accessing user profile information such as email ID, contact number, user choice or preference details, etc., for use in viral marketing. It requires a trust-based social network to diffuse information without affecting users' privacy. Prior information for collecting specific user data and using it further for viral marketing helps establish a trusting relationship between the company and users registered on its social network portal. It is essential to follow good marketing ethics based on respect for users' privacy, maintaining honesty, being transparent about rules and regulations, providing authentic data, and sharing genuine reviews/ feedback on products. In addition, user statistics like followers (visitors) count and user ratings must not be fake and misleading.

8 Basic terminology of influence maximization

The different concepts related to influence maximization are:

Definition 1 (Social Network) A social network is represented by a Graph G(V, E), consisting of nodes and

edges. The nodes of a graph are represented by set V, and the edges by set E.

Definition 2 (Influence maximization) On a given social network represented $G = (V, E)$, where V represents a set of nodes and E represents a set of edges. Influence maximization is the selection of k nodes from set V to maximize the overall spread achieved by k seed nodes.

Definition 3 (Seed Nodes) Seed nodes represent a starting set of nodes in an SN to start the information diffusion process. Usually, these are influential nodes, i.e., nodes of high importance, and such nodes help increase the influence spread represented by function $\sigma(x)$. They are assumed to be initial adopters of the information to be diffused and considered the only activated nodes at the beginning of the diffusion process. The selection of quality influential nodes is crucial for successful information diffusion. Disjoint sets represent seed sets for different competitors, and the nodes influenced by various seed sets also result in a non-overlapping set of nodes.

Definition 4 (Diffusion Threshold) It represents a value between (0,1] assigned to each node in an SN. It further signifies the strength of a node, and the node will be influenced if the influencer node(s) trying to influence it have more influential weight than its threshold.

Definition 5 (Diffusion Model) Researchers have introduced various information diffusion models to simulate diffusion. The diffusion takes place under specific rules defined by the model. The selection of appropriate diffusion models helps in maximizing the influence spread.

Definition 6 (Edge Diffusion Probability) In some diffusion models, a value between (0,1] is assigned to each edge in a social network, indicating whether a diffusion can be done through this edge. Usually, it is either similar for all edges or assigned randomly at the beginning of the diffusion process.

Definition 7 (Influence Spread) It is represented by $\sigma(x)$ function, which computes the number of influenced nodes at the end of the diffusion process for the seed set x. In a competitive scenario, this spread is calculated for each competitor, and one who gets the maximum spread value is declared the winner.

Definition 8 (Monotonicity) $\sigma(x)$ is the set function whose value increases when more nodes are included in the seed set, and it should follow the equation $\sigma(S) \leq \sigma(T)$, where S and T represent seed sets such that $S \subseteq T$. It means that the increase in influence spread is directly proportional to the number of seed nodes.

Definition 9 (Submodularity) It means when a set of seed nodes extends, its impact decreases in terms of influence spread, and we can refer to it as diminishing returns:

$$\sigma(S \cup \{x\}) - \sigma(S) \geq \sigma(T \cup \{x\}) - \sigma(T) \quad \text{Here } S \subseteq T \subseteq U, x \,\varepsilon\, U - T, \textit{and U is the universal set}. \tag{1}$$

In Eq. (1), S *and* T are sets that represent two different seed sets such that the number of seed nodes in set T is more than the number of seed nodes in set S and further $S \subseteq T$. The marginal gain in influence spread will be less when we keep increasing the number of seed nodes, and a situation arises where the nodes to whom one seed node is trying to influence are already influenced by another seed node. So, the count of newly influenced nodes will be less than the influence spread achieved by the previous seed set.

9 Types of influence maximization (IM)

Influence maximization basic framework includes the selection of a social network, selection of diffusion model, and seed set selection algorithm to increase the influence spread on the network. It can also be studied using different frameworks, such as user diffusion across multiple networks, context-aware IM, topic-aware and time-constrained IM, etc. Influence maximization research focuses on finding the seed nodes to maximize their aggregated influence spread. This kind of work has been carried out either at an individual level or at the community level. Individual IM always depends upon the topic.

Further, IM can be categorized as competitive and non-competitive IM. It can be solved using approximation algorithms, Heuristic and Metaheuristic Algorithms, and Community-based influence maximization algorithms. IM is an emerging field of research in dynamic networks. Seed selection is of utmost importance in solving IM, and it mainly uses three types of approaches: simulation-based, heuristic-based, and sampling-based algorithmic approaches (Singh et al. 2022). Identification of seed nodes can be done in one round or multiple rounds by utilizing the influence history, which also allows more flexibility and adaptability to changes. Figure 5 shows different types of IM.

9.1 Context-aware influence maximization (CAIM)

CAIM has emerged as a new dimension of IM, considering various context-related features such as topic, location, time, etc. Topic-aware influence maximization (TAIM) is studied as IM for topic-relevant targets, i.e., information is diffused to target topic-relevant nodes, and IM for topic-dependent where the edge between source to target node is topic-aware (Chen et al. 2016). In Topic-aware IM, the topic represents

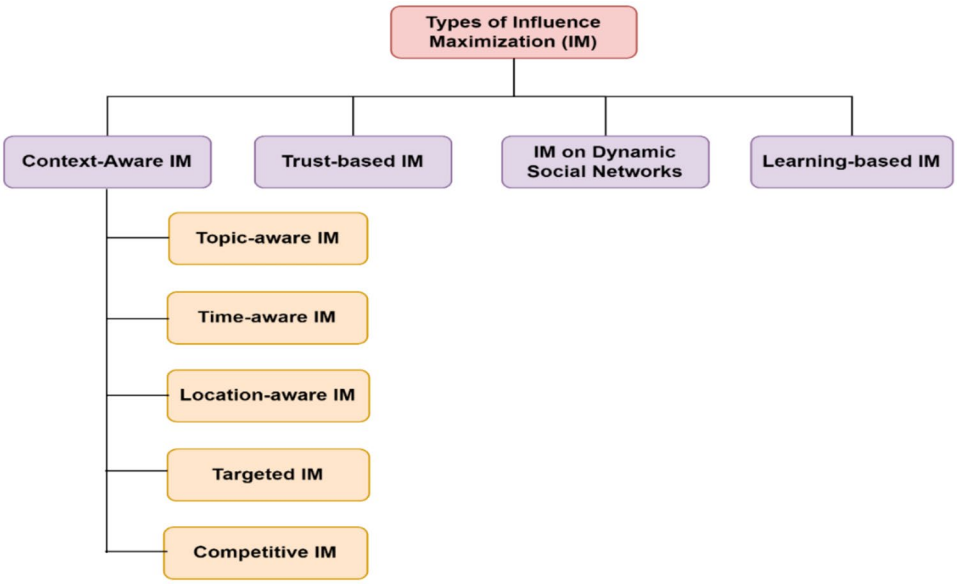


Fig. 5 Types of influence maximization

all those characteristics related to item and user interests. Diffusion occurs when seed nodes are topic-aware and target nodes are topic-relevant, i.e., nodes whose interest is similar to item characteristics. This kind of IM refers to targeted IM.

Time-aware IM takes care of time constraints for diffusion. The diffusion occurs either at discrete time steps or using "Continuous-Time diffusion models." In this type of IM, time-aware diffusion models, extensions of LT and IC models, are considered for simulating the diffusion process. Generally, diffusion occurs either at discrete time steps or on a continuous time basis. CT diffusion models are more practical in approach. So, Continuous-Time IC diffusion models are popular in time-aware IM (Li et al. 2018a).

Labeled or targeted IM refers to maximizing the influence spread among specific nodes, i.e., target nodes exhibiting particular characteristics. It is a special case of IM where seed node(s) selection is done to maximize influence for specific target nodes (Li et al. 2011) using the labeled social network IMDb. This variant of IM is more efficient in terms of getting more benefits by targeting a smaller number of relevant consumers, and it can also be extended to IM in multi-competitor scenarios (Carnes et al. 2007).

Similarly, location-aware IM considers IM for location-relevant nodes, i.e., information is diffused to location-relevant nodes on location-based social networks. In addition to this, IM on dynamic networks that are continuously evolving is also an emerging area of research. Further, competitive IM also comes under context-aware IM, where more than one competitor tries to diffuse information to increase the influence spread in competition to their opponents. The opponent's strategy is either known to another competitor beforehand or may not be known to the other competitor.

9.2 Trust-based influence maximization

In trust-based IM, the ID takes place under the assumption that a node influenced once by any competitor will not change its decision till the end of the diffusion process. Further, another variant of trust-based IM can be based on signed networks such as Slashdot, where the sign of the edge is either positive or negative. A negative edge label indicates that diffusion across that edge is not feasible as it represents a negative relationship (foe), and diffusion takes place only through edges labeled with a positive sign (Zhuang et al. 2013), which means a positive relationship (friendship).

9.3 Influence maximization in dynamic social networks

Information diffusion (ID) on Dynamic Social Networks is one of the emerging areas where modeling ID is more complex than static network-based diffusion. In a practical scenario, online social networks keep changing in terms of the number of nodes and relationships between them with changing times, which directly impacts influence dynamics (Zhuang et al. 2013). The scalability of networks and changes in interconnections may also change the influential node's status; that is, its strength of influence will also be affected.

9.4 Learning-based influence maximization

To address the drawbacks of traditional methods, like lack of generalization ability and difficulty in finding optimal seed sets due to the stochastic nature of ID, researchers have studied deep-learning methods for IM in the past decade

(Ling et al. 2023). The significant challenges faced by leaning-based IM methods include how to solve the objective function in a better way, characterization of the diversified underlying diffusion patterns, and the difficulty of adapting the solution under various "node-centrality-constrained" IM variants. A novel framework, DeepIM, was proposed, and a novel objective function was designed to resolve the difficulties (Ling et al. 2023).

10 Competitive influence maximization (CIM) and its applications

In a competitive scenario, the IM problem is referred to as CIM, where there is more than one competitor, and it can be done at the whole network or the community level for better performance. Competitive research for IM is based on information to be spread for single or multiple items/technologies. This research problem is also studied by researchers where one competitor tends to stop the other competitor from influencing nodes in the network. The overall goal is to increase the influence spread (IS) of influential nodes to the maximum extent, whether we know the competitor's strategy or not. In case if another competitor's strategy is not known, then the model should use a "learning-based algorithm" for adapting to changing situations. This kind of research is in high demand in the business and marketing industry. The aim is to maximize the IS for each competitor, and the winner will be that competitor who will influence the maximum nodes on the social network. In the generalized CIM problem, the problem is formulated as:

The number of competitors say N_c and a set of competitors, $C = \{C_1, C_2, C_3, \ldots .., C_n\}$

Seed sets for each competitor are represented by set $S = \{S_1, S_2, S_3, \ldots \ldots, S_n\}$

Here S_1 refers to the seed set of a competitor, C_1, S_2 refers to the seed set of competitor C_2, and so on.

Number of seed sets (N_s) = Number of competitors (N_c) i.e

$N_s = N_c \; or \; |S| = |C|$

The number of seed nodes assigned to each competitor can be kept the same or different as per the requirement of research problem formulation. Seed nodes are conventionally allocated based on the follower or host perspectives. Seed nodes may vary in influence strength according to their influence score or rank. From a budget constraint perspective, influential nodes or seed nodes are assigned as per the budget available with different competitors for hiring influencers to market their products or information. For simplicity, the budget can be considered the same for all competitors, and now, the main issue will be assigning seed nodes fairly to all competitors. Figure 6 shows each competitor's seed node allocation and influence spread computation.

Competitive influence maximization has applications in viral marketing, election campaigns through influential node identification, etc. Viral Marketing has been studied in competitive and non-competitive scenarios. In a competitive scenario, more than one company is involved in advertising their product/technology information such that all try to achieve maximum influence spread in competition with their opponents. A company's success is highly dependent upon efficient viral marketing strategies.

CIM is also highly used for event management, and specific portals are designed to propagate information about events such as conferences, workshops, training programs,

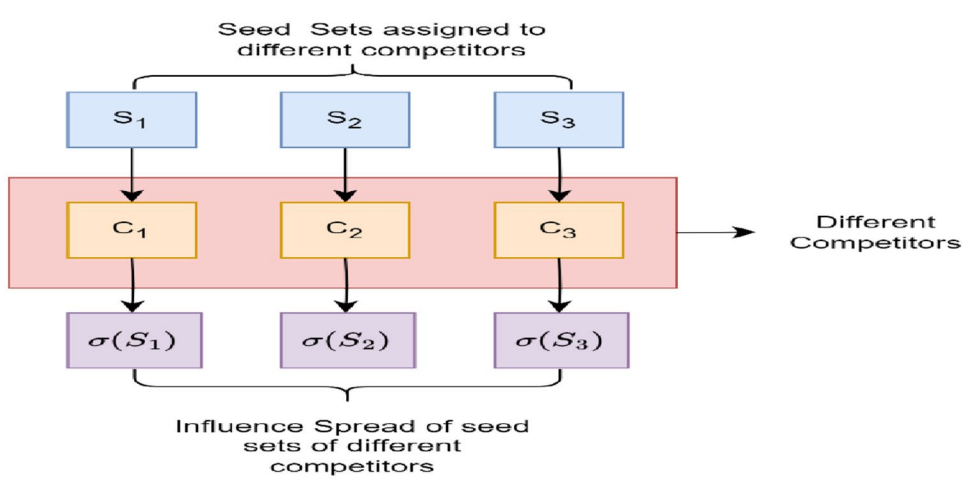


Fig. 6 Seed allocation to each competitor and mapping with influence spread

etc. The success of training programs in online and offline modes of short/ long duration depends upon a good number of admissions, which results from the efforts of influencers and the budget assigned for online marketing.

Election campaigns are also considered applications of CIM because multiple political parties want to spread information about their party achievements, policies, and agenda for the upcoming elections. It is one of CIM's most popular application domains, and a lot of money is spent on hiring influencers to promote political parties' agendas for the upcoming elections.

11 Information diffusion models for competitive and non-competitive influence maximization

Influence study or influence analysis is vital in predicting information diffusion based on user behavior analysis and network structure. The information diffusion process involves complexity due to different types of social networks varying from static to dynamic, homogeneous networks to heterogeneous networks, and sparse to dense networks. Further, many users in the network with different preferences, and various constraints related to social networks also affect information diffusion (Tejaswi et al. 2016). This paper mainly focuses on two categories of information diffusion models namely competitive and non-competitive diffusion models. When diffusion of information is to be done concerning more than one company (competitor) then it becomes a competitive ID.

ID models can be either "*progressive or non-progressive*" (Masucci and Silva 2014). Most of the IM algorithms make use of progressive diffusion models. In progressive diffusion models, a node that once becomes active will remain active throughout the diffusion process. The diffusion process continues till any node in the network keeps on becoming active. Once no such activity is observed, it indicates that no further diffusion has occurred in the social network. In non-progressive diffusion models, nodes activated once can also be de-activated again. The information diffusion process, in general, is stochastic, i.e., it exhibits randomness.

11.1 Information Diffusion Models for non-competitive scenario

The models mostly used for influence maximization are the LT Model, IC Model, Game Theory Model, Triggering Model (TR), and CT Model (Continuous-time Model). LT and IC models can be used in both scenarios, i.e., for competitive and non-competitive diffusion. In a competitive scenario extended version of classical models can be used.

The LT Model is threshold-based and is usually called a receiver-centric diffusion model. In this model, the probability of being influenced keeps increasing with the activation of its neighboring nodes. It is a special case of the "Generic Threshold (GT) Model". The total weight of the edges from the neighboring active nodes of the node says 'v' should be less than or equal to 1 (Lu and Bonchi 2013). Further, every node, say v, takes a random number from [0,1] as its threshold value.

In the IC Model, every active node is allowed once to activate all the neighboring nodes that are inactive. If more than one active node is trying to activate an inactive node, then they are given a chance to activate it in arbitrary order. Further, each edge is associated with a probability, i.e., constant or randomly assigned value (weight) according to some criteria. This model is also known as the sender-centric model. A node activated once remains active from that iteration onwards, and further, it tries to influence its neighboring nodes (Ansari et al. 2019) (Li et al. 2018b). The probability of diffusing information is the same on all the edges of the SN. This model is used to achieve IM but it is incapable of capturing influence propagation delay. The propagation process stops when no new node becomes active, as in the case of threshold-based models (Dai 2018). There are different variants of the IC model, like the WC model, where the probability of node u trying to influence node v varies, and the probability of diffusion along each edge can be set to $1/d_v$ (Chen and Wang 2009) or random weights can be assigned to different edges. Further, the Latency Aware Independent Cascade Model (LAIC) takes care of latency information about information (influence) diffusion in the SN (Jayamangala and Sheshasaayee 2015).

LT and IC Models are considered special cases of the Triggering (TR) Model (Masucci and Silva 2014). TR Model was proposed to generalize the LT and IC Models and make use of distribution for mapping a subset of neighboring nodes of any node, say 'v' to a probability such that these nodes can influence node v. LT, IC, and TR Models possess both the characteristics namely submodularity and monotonicity.

In Heat Diffusion Model, the nodes representing hubs are considered initial nodes from where the diffusion process starts and spreads further. These nodes act as a heat source, and the diffusion process is similar to the heat diffusion process. This model is also not suitable for dynamic networks.

Researchers have tried various extensions based on the above ID models

The more generic form of the IC Model is the "*Decreasing Cascade Model*" (Zheng 2018), where the probability of a node becoming active reduces when many of the neighboring nodes have already attempted to activate it. In the IC Model with negative opinions, the quality factor helps represent the real-time situation where users accept the negative

opinions corresponding to product deficiency. It makes every node in the network either positively active with probability, say q_1 or negatively active with probability $1 - q_1$.

G. Tong et al. (Tong et al. 2017) proposed the Adaptive Seeding Strategy(ASS) using the extended IC model called *DIC Model* to take into account the uncertainties of the diffusion process, such as the seed users may not be surely activated if the seed nodes are not satisfied by the sample product to be advertised further through them. So, the idea for proposing the DIC model is to seed the user more than once if not activated in the first round. Hence, it involves considerable efforts to activate a powerful, influential node in the network that may generate a great influence on the social network. Another kind of uncertainty can be the dynamically changing structure of SN in terms of "degree and relationships" between nodes. In the DIC model, the propagation probability is a random variable that supports the requirements of dynamic SN.

A. Goyal et al. (Goyal et al. 2010) proposed an algorithm and a model for learning its parameters to test the trained model for predicting the time by which the user may start acting, following its neighboring nodes. In this research, the authors mainly focused on mining the influence probabilities based on past influence cascades (action logs), i.e., users' past behavior. Action logs are represented as A (User, Action, Time), i.e., "**(u, a, t_u)**" which indicates that user "***u***" performs an action "***a***" at a time "$\mathbf{t_u}$." For capturing the influence, the authors have proposed static and time-dependent models, which showed a reasonable performance. Out of all the proposed models, the CT model is the most expensive and has shown the best performance. The performance of the DT model is also almost identical to the CT model in terms of efficiency and accuracy. The authors have adopted an instance of the GT Model as a framework, and individual influence probabilities between users are assumed to be constant, as we generally find in viral marketing. Still, in the future, we can also assume dynamic influence probabilities.

11.2 Competitive influence diffusion models

Carnes et al. (Carnes et al. 2007) proposed models, namely the "Distance-based Model and Wave Propagation Model," to spread the influence of competing technologies through SN and solve the IM problem by using the follower's perspective. In this approach, the set of consumer nodes for the first competitor is already known, making it easy for the second competitor to select the seed nodes with more IS within a smaller budget. The models used by the authors are the versions of the IC model for the CIM scenario. The results achieved by the authors can be modified to handle more than two competitors and different probabilities of acceptance can be considered for the two technologies.

In the "Distance-based IC model," each edge is represented by $(u, v) \in E$, mapped with a length d_{uv}. In the absence of length information, all the edges are assigned with length one or any arbitrary non-negative value. The initial set of adopters for technologies say A and B are referred to as I_A and I_B such that total initial adopters are represented by a set, $I = I_A \cup I_B$, and $I_A \cap I_B = \varnothing$. Let $d_u(I, E_a)$ refers to the shortest distance from the node u to I along the active edges, E_a of the network. So, the node will be influenced or adopt the technology of the initial adopter who is nearest to it. Moreover, $d_u\left(I, E_a\right) = \infty$ if and only if u is not connected to any node of I using only active edges. It means u does not adopt any technologies out of A and B.

Xinran et al. (He et al. 2012) studied CIM using the CLT Model to solve the "Influence Blocking Maximization (IBM)" Problem. The solution exhibits submodularity, and hence, the approximation ratio attained using a greedy algorithm is $1 - 1/e$. It is not as efficient as it uses Monte Carlo simulations of competitive influence propagation. Every node is in mainly three states, namely "inactive," "-active," and "+active." Once the node becomes active, it remains active. Negative and positive influence propagates in the LT model, and the node becomes active according to the influence that dominates, i.e., positive influence or negative influence.

Wei Lu et al. (Lu and Bonchi 2013) solved the CIM problem in the field of VM from the host or SN owner perspective, and the host was allowed to select the seeds for each competitor. The authors proposed a propagation model named K-LT to capture the nature of competitive VM and satisfy "*submodularity and monotonicity.*" In this model, node activation takes place in two phases. A node gets influenced by its active in-neighbors irrespective of a specific competitor, and its state changes to an active state. In the second phase, the node determines which competitor's incoming influence is more (Lu and Bonchi 2013).

A. Borodin et al. (Borodin et al. 2010) discussed WPCLT and STM for competitive influence maximization. In the WPCLT model, a threshold value θ_v is assigned to each node initially from [0, 1] and every edge (u, v) belonging to Graph G is assigned a weight $W_{u,v}$ from [0,1], and it indicates the influence of node u on node v. At any time instance, node $v \epsilon V$ in the graph G, is influenced by competitior1, if and only if the ratio of the sum of weights of all the incoming edges coming from active nodes of competitor1 and the total weight of all incoming edges from all active nodes is more than θ_v. Similarly, influence can be analyzed using this model for any number of competitors in a multi-competitor environment. The node will be influenced by a competitor whose influence ratio has a maximum value and more than the threshold value of a node, v. Further, the WPCLT model considers all the active neighbors in the previous time steps, and this helps in considering old and

recently activated nodes to influence another node, but on the other hand, the K-LT model at time step t, only depends on recently activated nodes of the time step, $t-1$. In STM, the difference is in terms of separate threshold values for each node in graph G are assigned for each competitor. It means multiple threshold values, say $\theta_A, \theta_B, \theta_C$, etc. are assigned to a node if there are three competitors say A, B, C. In previous LT based models, same threshold values is set for all competitors. This assumption gives more flexibility to the consumer to adopt a technology.

A. Bozorgi et al. (Arrami et al. 2017) focused on solving the CIM by introducing a new diffusion model called DCM based on the LT Model. In this model, a node can have three states, i.e., inactive, *active*+, *active*− and *thinking state*. Here *active*+ indicates nodes influenced by *competitor*1 represented by + sign and *active*− indicates nodes influenced by *competitor*2 represented by—sign. Initially, only the seed nodes selected or assigned to Competitor 1 and Competitor 2 were in active state, and the remaining nodes were inactive. Each node in the network is assigned a threshold value, θ_v from [0,1]. The status of node v changes to the thinking state if the influence from the active in-neighbors of v is greater than or equal to its threshold value. The node will be in a thinking state till $\varepsilon d\varepsilon$ time steps, and after that, it will have to decide to become active by *competitor*1 or *competitor*2 based upon whose influence on v is more.

(Pham et al. 2019b) proposed the TCLT model by extending the CLT model to simulate influence maximization in a competitive scenario within a time constraint t. The problem is formulated for two competitors say X and Y, who want to advertise their products on OSNs under the assumption that the seed set for competitor Y is already known to competitor X and the budget 'B' available is limited, and each node is assigned with a heterogeneous cost to be activated by competitor X.

Y. Dai et al. proposed a CG-LT model based on the LT model for solving the GCIM problem. A node v in a community (group), say C_i is activated if and only if the number of activated neighboring nodes of a node v are $\geq \rho *$ neighboring nodes of a node v(ρ is an influence factor between 0 and 1). After a decision is taken by nodes in a community, the overall decision of that community is based upon the number of activated nodes in a community i should be $\geq \rho *$ number of nodes in a community. So, an influenced group tries to influence its neighboring groups who have not yet taken a decision. The influence factor decides how fast the influence spread will take place (Dai 2018).

H. Huang et al. proposed the CCIC diffusion model (Huang et al. 2021). It is similar to the IC model in the diffusion process but differs regarding multiple entities (competitive and complementary products) propagating in the SN. The nodes can be activated by various products sequentially. Another difference with the IC model is that probabilities of influence between two nodes on different target products are different.

LT, IC, and GT models are also called predictive models or influence-based models. These models predict ID (Li et al. 2018a), find influence spread, and maximize influence. Generally, Game Theory Models are more suitable for predicting ID in dynamic SN, and they should be robust enough to meet the changing dynamics of SN and interaction among nodes. The existing models for the prediction are classified as Graph and non-graph-based approaches. The IC and LT models use the static graph structure, i.e., a directed graph underlying the diffusion and activation of each node depending upon the monotonicity assumption. Moreover, the nodes that are activated once cannot be deactivated.

12 Competitive influence maximization (CIM) variants

Many researchers have studied IM as a non-competitive environment. Still, in applications like e-commerce, election campaigns, event management, education, and training sectors, CIM plays a significant role in spreading messages, innovative ideas, and product information more effectively than its competitors. In general, the CIM problem refers to the spread of influence by multiple competitors over social networks to get maximum influence spread compared to competitors. The different variants of CIM are:

12.1 Competitive influence maximization at community level

It is preferred over the whole network due to the less time complexity involved in identifying influential nodes on SN. In this approach, as explained in Fig. 7, communities are identified in SN using efficient algorithms. Secondly, select the appropriate information diffusion model for CIM. Thirdly, there is a need to find top-k influential nodes at the community level and assign them to all competitors using the follower or host perspective so that the overall influence spread of each competitor can be maximized. Seed nodes can also be assigned randomly or using a game theoretic approach to each competitor. Finally, the competitor with the maximum spread will be announced as the winner.

12.2 Influence blocking maximization problem

Competitive influence maximization is studied mainly as a problem of maximizing the influence of one competitor over another or as a problem of minimizing the influence of other competitors (Dai 2018).

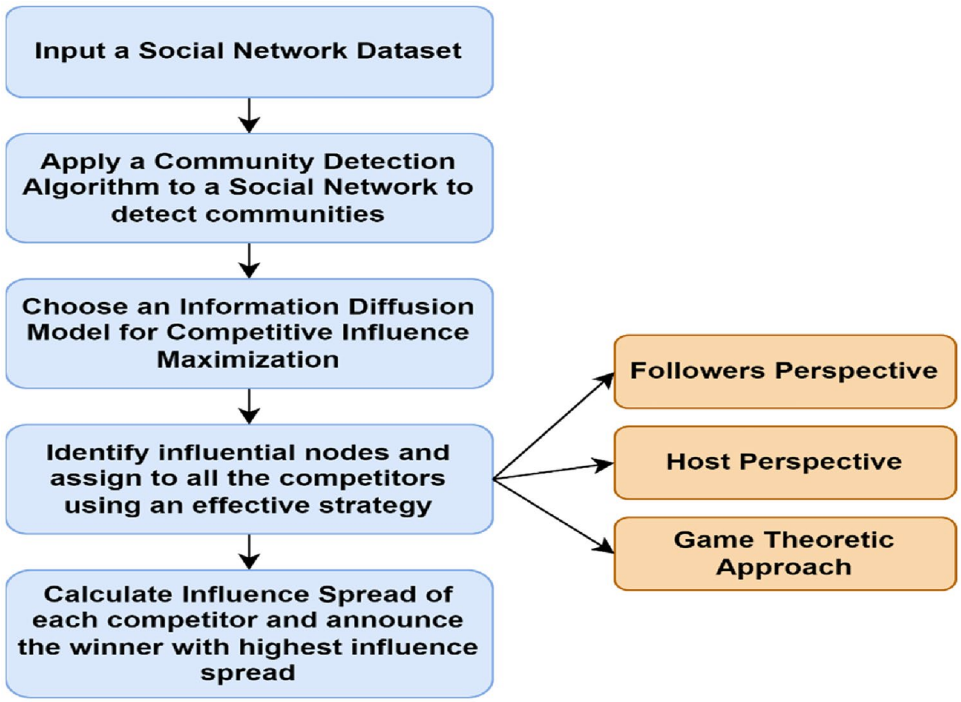


Fig. 7 Flow of competitive influence maximization at the community level

12.3 Fair seed allocation (FSA) or host perspective

Researchers study the variants of CIM, and one of them is FSA, where the goal is to assign seeds to all competitors in a fair manner by the host of the SN. The various approaches used for assigning nodes, like randomly assigning the seeds to all the competitors, using heuristics like ordering nodes as per the highest degree centrality, and then randomly selecting a competitor to whom nodes at priority can be assigned, etc.

12.4 BCIM (Budgeted Competitive Influence Maximization)

BCIM refers to allocating seed nodes to a competitor, say A, assuming that competitor $\boldsymbol{A}$ already knows about seed nodes assigned to competitor $\boldsymbol{B}$using limited budget $\boldsymbol{B'}$ and time constraint t. Further, the cost with which each node can be influenced is heterogeneous. The diffusion model TCLT is used, which takes care of time constraints and is an extension of the CLT model (Pham et al. 2019b).

12.5 CPro (Competitive Profit Maximization for the host)

This CIM problem variant deals with maximizing the profit achieved by the host of the online social network owner in a multi-competitor scenario. Multiple campaigners (competitors) are doing viral marketing of their respective products using the same OSN owned by the same host, and each campaigner pays profit (commission) to the host if a node in the network adopts their product (Zhu and Li 2018).

12.6 CIM Method considering Inactive Nodes and Community Homophily (CIMMIC)

In this method under the competitive information propagation model considering inactive nodes (CIPMCIN), minimum number of the most influential seed nodes with maximum spread are preferred for an *competitor* $C2$ against *competitor* $C1$. The community boundary node influence on other communities is computed, and a node having maximum influence (acting as a bridge) is used for selecting the seed node of *competitor* $C2$ till it becomes more than the influence of competitor $C1$.

12.7 GCIM (Group-based Competitive Influence Maximization)

GCIM focused on identifying high-performing seed sets in terms of minimum average activation cost in an influence spread (Dai 2018). The two types of effects are introduced to increase the influence spread, i.e., intra-group and inter-group. The nodes in the same community try to influence nodes in their community, and the final influence for a competitor out of multiple competitors is selected on a majority basis. After deciding on the community (group) level, the group decision is used to influence other communities (groups) in the SN, and this also refers to the inter-group effect. Influence factor (ρ) having a value between 0 and 1 is

used while influencing nodes in a particular community and influencing another group by an already influenced group. In addition, the higher the influence factor value, the less the influence of local effect on group-based IM will be.

12.8 User-driven Competitive Influence Maximization

This variant of CIM is known as the User-Driven Competitive Influence Maximization (UDCIM) problem. In this CIM variant, the issue of differences in user preferences for the products under the same category is addressed by introducing the User-Driven Competitive Linear Threshold (UDCLT) model. Researchers also introduced a community structure-based novel measure, Topology Importance (TI), to assess a node's potential influence within a social network by considering its connections within and across communities. In addition, they developed a novel two-phase algorithm, "Community-based Dual Influence Assessment (CDIA)," to resolve the UDCIM problem by integrating Topology Importance and Dual Influence to identify seed nodes (Liu et al. 1018).

12.9 Competitive influence maximization in trust-based social networks with deep Q-learning

Deep reinforcement learning methods can be applied to estimate the influence strength and train a controller for selecting a node list of a given length as the initial seed set for the information spread. The deep Q-learning methods are suitable for analyzing the CIM on trust and distrust-based social networks (Kopacz 2024).

13 Related work on competitive influence maximization

Table 3 describes work done by various researchers in the field of CIM in the last few years. The most commonly used approaches were based on greedy algorithms or their improvements, approximation algorithms, and greedy heuristic algorithms to identify seed nodes for initiating diffusion on social networks. Efficient IM is highly dependent upon relevant (significant) seed nodes, and this selection may also vary according to the domain of IM. Further, top-k seed node selection directly affects ID, and many researchers have also suggested various influence measurement scores that lead to the maximum spread of ideas or information on the SN. Choosing an appropriate diffusion model in a competitive scenario also plays a vital role in the ID process. The increase in the number of seed nodes need not contribute more towards an increase in influence spread or marginal gain, i.e., monotonicity and submodularity are not guaranteed to be achieved. The main goal of IM is to increase maximum spread by selecting an optimal number of seed nodes and efficient budget utilization under time constraints. The seed nodes in a competitive environment are often chosen using follower or host perspectives. The influence dynamics become more complex over dynamic networks and changing behavior of nodes on the network. The nodes activated earlier by one competitor may change their preference for another competitor's influence in the future. So, CIM has become popular in the last decade due to the marketing preference of companies using social media platforms that are more economical, quicker, and less time-consuming.

14 Datasets for competitive influence maximization

Most of the datasets used for the research on CIM are from the Stanford University website (“Stanford Large Network Dataset Collection”. 2021), and their details are given in Table 4. In addition, datasets can also be downloaded from the Kaggle website (“Kaggle Social Networks Datasets”. 2024) and graph and network repository(“Graph and Network Repository”. 2024). The datasets in Table 4 are of different types: Directed, Undirected, Undirected Unweighted, and Directed Weighted. Further, datasets vary from medium-scale to large-scale social networks, so the results can be verified for scalability.

In addition to the social networks in Table 4, many other social network datasets can be downloaded from the network repository (“Networks Repository”. 2022).

15 Conclusion and challenges of competitive influence maximization

CIM is one of the popular variants of IM due to its increasing applicability in the viral marketing domain, targeted marketing, or recommending as per personal or group preferences. There is a focus on two approaches for seed selection for different competitors: the followers approach and the host approach. Most of the work done in the direction of CIM is based upon greedy heuristic algorithms for finding efficient seed sets for competitors. However, there is still scope for faster and more scalable algorithms for seed node identification. In most research studies, the objective function for IM exhibits the properties of “monotonicity” and “submodularity,” but this is not guaranteed in all works. The diffusion models for CIM are based upon LT and IC models with modifications for time and budget constraints

Table 3 Literature survey on competitive influence maximization

Algorithms used (Ref. No., Year)	Approaches used (objective function)	Influential nodes selection strategy (competitors strategy)
Greedy hill climbing algorithm Carnes et al. (2007)	Approximation algorithm (monotone and submodular)	High-degree nodes and central nodes based on low average distance from other nodes (follower perspective for two competitors)
CLDAG Bharathi et al. (2007)	Greedy approximation algorithm	Negative seed set and positive seed set are fixed (Influence Blocking approach used by one competitor to block another competitor)
Needy greedy (efficient and scalable) Lu and Bonchi (2013)	Greedy heuristic algorithm (monotone and submodular)	Seed node allocation from the Host Perspective was done to maintain fairness in seed allocation for different competitors having different budgets to hire seed nodes (host perspective)
Speaker-listener label propagation (for community detection) and CI2 algorithm for finding influential nodes Lu and Bonchi (2013)	Greedy algorithm	Seeds allocated to the first competitor were known to the second competitor, and seed nodes for the second competitor were found using the CI2 algorithm (follower perspective)
DCIM_CELF algorithm (low efficiency for scalable large-scale networks) Li et al. (2018b)	Greedy algorithm (monotone and submodular)	In the CELF algorithm, the initial seed set is selected based on a node with the highest influence spread (follower perspective)
Scalable approximation algorithm Zhu and Li (2018)	Approximation algorithm (Non-monotone and non-submodular)	This algorithm gives better performance over existing techniques for profit maximization, even for billion-scale networks like Twitter (host perspective)
Select group in groups (SGG) greedy heuristic algorithm to identify high-performing seed set improved louvain algorithm (to detect the communities) Dai (2018)	Greedy algorithm	Improved greedy algorithm with page rank algorithm is used to select seed nodes in a group
"Sandwich approximation based on polling-based approximation (SA-PBA)" (effective and scalable) Pham et al. (2019b)	Approximation algorithm (neither submodular nor supermodular)	Budgeted CIM problem is dealt with using the TCLT model under time and budget constraint
"Deep recursive hybrid model (DRH) and an greedy influence maximization approximation algorithm (DRHGA)" Huang et al. (2021),	Approximation algorithm	The proposed algorithm proved to be efficient for finding the seed nodes set for the target product as compared to other methods in terms of predicting accuracy and efficiency (follower perspective)
Two-phase seed node selection algorithm (SNSA with CB) Xie et al. (2021)	Greedy algorithm with community boundary influence	The CIM method based on inactive nodes and community homophily called CIMMIC is compared with the CI2 Method and proved better in efficiency and feasibility (follower perspective)
Loyality based competitive influence maximization (LBCIM) with Ɛ-greedy MCTS(monte-carlo tree search) strategy Alavi et al. (2021)	Greedy algorithm	A seed node for each competitor is selected as per turn, out of inactive nodes and nodes influenced by an opponent party. Here node activated once by one competitor may get influenced by another opponent by offering better benefits

Table 4 Datasets description for competitive influence maximization

S. No	Dataset name	No. of nodes and edges Type of graph	Description
1	Arxiv HEP-TH "Stanford Large Network Dataset Collection- Arxiv HEP-TH". (2022)	27,770 nodes representing papers and 352,867 edges Directed	High energy physics theory citation network, if paper k cites paper l, then there is an edge between nodes representing papers k and l
2	Ca-HepPh "Stanford Large Network Dataset Collection–Ca-HepPh". (2022)	12,008 nodes and 118,521 edges Undirected	This network is based on scientific collaborations between authors' papers submitted to the "High Energy Physics—Phenomenology category." The undirected edge from "i " to " j " means author " i " co-authored a paper with author " j ". If, in case, the paper is co-authored by "k " authors, then it results in a completely connected subgraph on "k " nodes
3	Epinions social network "Stanford Large Network Dataset Collection–Epinions Social Network". (2022)	75,879 nodes and 508,837 edges Directed	Epinions is a "who-trust-whom" OSN of a general consumer review site Epinions.com. The site members can decide whether to "trust" each other or not. The interactions based on trust relationships result in the Web of Trust, combined with review ratings, which determines which reviews will be shown to the user
4	Flixster "Flixster online social network". (2022)	2,523,386 nodes and 7,918,801 edges Undirected and Unweighted	This OSN, based on movie rating sites, helps people meet people with similar movie tastes. Here, the node represents the user, and the edge represents friendship
5	Slashdot "Slashdot social network". (2022)	82,168 nodes and 948,464 edges Directed	This network contains friend/foe links between users of Slashdot (a news website related to technology)
6	Amazon "Amazon co-purchasing network with ground truth communities". (2022)	334,863 nodes and 925,872 edges Undirected	Amazon is a co-purchasing network. An edge from any node "u" to "v" represents items co-purchased by a customer
7	Wikivote "Wikipedia vote network". (2022)	7115 nodes and 103,689 edges Directed	The network contains directed edges, representing node i (user i) voted on node j (user j)
8	Facebook_combined "Social Circles: Facebook". (2022)	4039 nodes and 88,234 edges Undirected	The dataset includes friendship lists from Facebook. Edges from all ego-nets are then combined to get the dataset, which helps identify users with the same political affiliations
9	DBLP "DBLP Collaboration Network and Ground-truth Communities". (2022)	317,080 nodes and 1,049,866 edges Undirected	DBLP is a co-authorship network, where an edge represents the nodes (authors) who have published at least one paper in collaboration
10	NetScience "Networks Repository". (2022)	379 nodes and 914 edges Undirected	Nodes in this collaboration network represent researchers, and an edge represents a co-authorship relationship
11	CaGrQc "General Relativity and Quantum Cosmology Collaboration Network". (2022)	5242 nodes and 14,496 edges Undirected	This collaboration network contains undirected edges between nodes representing authors who co-authored a paper

for competitive scenarios where more than one competitor wants to spread influence on the OSNs.

Further, the CIM problem has been studied by various researchers in different forms, namely BCIM, DCIM, FSA, LBCIM, CPro, etc. The survey includes multiple factors that affect the diffusion process like the structure of the network, seed node selection, evolving nature of online social networks, changing behavior and choices of users, company marketing strategies, budget and time constraints, network size, etc. which are essential to understand before modeling ID models for the competitive scenario. The success of CIM depends upon the proper selection of the diffusion model, fair allocation of seed nodes under budget constraints, time-dependent influence analysis, competitive offerings by various competitors, and trust-based influence diffusion. The research work done in the area of CIM is less, and the survey findings suggest that there is a great need to explore the aspects and challenges as listed:

1. The CIM work is limited to two competitors but can be extended to more than two competitors.
2. Influence propagation can be studied using new variants of influence propagation models for competitive scenarios based on LT and IC models.
3. The community-based CIM can also be explored for overlapping communities, having nodes acting as a bridge between two communities.
4. This field still has the scope to work in the direction of faster seed selection algorithms under budget constraints at the community level to reduce time complexity or adopt better heuristics.
5. Another critical challenge in CIM is to meet the properties of Monotonicity and Submodularity.
6. Scalability is another important challenge to meet so that the CIM approach can work with large-scale real-world SNs, and its performance should not be limited to only small and medium-scale networks.
7. Robustness is also significant in identifying seed nodes if there is a change in information diffusion or propagation process.
8. Researchers can explore the work on CIM for dynamic social networks.
9. Community-based structures can play a vital role in identifying seed nodes that have more influence not only within their community but also outside the community. Metrics such as community density and participation coefficient can help rank nodes at the community level so that seed nodes (influential nodes) can be chosen in a better way to increase influence spread across communities. There is an excellent opportunity to design new ranking heuristics for efficient seed node selection.
10. Moreover, border nodes or boundary nodes having one or more edges connecting them with nodes in other communities can increase influence spread significantly.
11. In addition to the follower and host approach for influential node selection, the game theoretic approach offers tremendous possibilities for fair seed selection and allocation among various competitors.
12. Another essential challenge is handling the overlapping influence of competitors, which refers to a situation where more than one competitor tries to influence the same node in the network simultaneously. This situation needs a tie-breaker in terms of the influencer who dominates.
13. Context-sensitive influential node selection and budget constraints can bring a more realistic solution to the CIM Problem.
14. IM algorithms were mainly designed to solve a particular type of IM problem, but IM strategies can also be worked out according to the type and structure of network datasets.

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